Computer Organization
Homework 5

1. The Average Memory Access Time equation (AMAT) has three components: hit time, miss rate, and miss penalty.For each of the following cache optimizations, indicate which component of the AMAT equation is improved.（25%）
   (1) Using a second-level cache
   (2) Using a direct-mapped cache
   (3) Using a 4-way set-associative cache
   (4) Using a virtually-addressed cache
   (5) Performing hardware pre-fetching using stream buffers
   (6) Using a non-blocking cache
   (7) Using larger blocks

2. Design a 128KB direct-mapped data cache that uses a 32-bit address and 16 bytes per block.(1 word=4 byte)Calculate the following: （25%）
   (a) How many bits are used for the byte offset?
   (b) How many bits are used for the set (index) field?
   (c) How many bits are used for the tag?

3. Consider a 256-byte 2-way set-associative write-back cache with 16-byte blocks. Assume that the cache does not store any data at the beginning, and the least recently used replacement policy is used. Consider a sequence of memory references whose byte address (in base 10) are 0, 1,256,128,and 0. The type of each memory reference (ordered from left to right) is indicated below. （25%）

Address	0	1	256	128	0
type	write	write	read	read	read

   (a) Determine whether there is a cache miss or hit for each memory reference.
   (b) Determine whether a write-back to main memory occurs for each memory reference.

4. Assume the three caches below, each consisting of 16 words. Given the series of address references as word addresses: 2, 3, 4, 16, 18, 16, 4, 2.
Pleaselabel each reference as a hit or a miss for the three caches (1), (2), and(3) below. Assuming that LRU is used for cache replacement algorithm and all all the caches are initially empty. （25%）

(1) a direct-mapped cache with 16 one-word blocks;
(2) a direct-mapped cache with 4 four-word blocks;
(3) a four-way set associative cache with block size of one-word.

address	Block No.	cache tag	cache index	H/M